

Norm Matloff's Debugging Tutorial

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(Please mail any questions to [Norm Matloff](#).)

Why waste huge amounts of time debugging? **Do yourself a favor and learn to debug like a professional.** In this Web page I have some resources which hopefully will be useful to you in that regard.

- **Our featured item -- a slide show!!** [Click here to download.](#) **Note that the point is NOT the technical details of DDD, but rather the general principles of debugging.**
 - An introduction to the secret art of debugging. Shows you the basic principles, leading you step-by-step in applying those principles to debugging two sample programs.
 - Click here if you wish to download the sample programs [Ins.c](#), [InsLotsOfErrors.c \(buggy version of Ins.c\)](#) and [Lnk.c](#).
- Information on some specific debugging tools:
 - My selection criteria:
 - Free.
 - Advanced features (e.g. display of linked data structures).
 - I prefer standalone debuggers, not integrated development environments (IDEs) --- I want to use my own favorite text editor -- but these IDEs are good for those who prefer IDEs.
 - Standalone debuggers:
 - DDD: For C/C++/Java (version 3.3)/perl/etc. UNIX. Featured in the [slide show](#) cited above. [Detailed information here.](#)
 - GVD: For C/C++. UNIX/Windows. Very much like DDD, but trimmer, faster-loading. [Detailed information here.](#)
 - JSwat: Excellent debugger for Java. Works on most platforms. Can be used as standalone or with the JIPE IDE (see below). [Detailed information here.](#)
 - IDEs: KDevelop, Eclipse, NetBeans etc.
 - Good use of a powerful text editor can speed up the development/debugging process. Links:
 - My tutorial on [general editing tips for programmers](#).
 - Vim: Very advanced extension of vi. Very popular. Multi-platform. Free. [Detailed information.](#)
 - Emacs: A classic. Highly programmable. Multi-platform. Free. [Detailed information.](#)
 - JEdit: Beautiful product. Great features. For C/C++/Java/etc. Multi-platform. Free. [Detailed information.](#)
 - [Links related to debugging.](#)

Other software Web sites by Norm Matloff:

- Professor Matloff's [Unix/C tutorials](#).
- His [Java tutorial](#).
- Dr. Matloff's [beginner's guide to installing and using Linux](#).
- His ["Extremely Quick and Simple Introduction to the Vi Text editor,"](#) and his introductions to the [elvis](#) and [vim](#) extensions to vi. The latter are much better than ordinary vi (and far, far better than pico). X11 mouse capability; subwindows (even on nongraphics terminals); infinite undo; paragraph formatting and so on.
- Norman Matloff's [guide to fast debugging!](#) See also his [text-editing tips designed especially for programmers](#).
- Professor Matloff's [LaTeX tutorial and resource page](#), and his [tutorial on LyX](#), a GUI interface to LaTeX.

- Dr. Matloff's introduction to [the mutt e-mail utility](#). Don't use pine! It was designed for people who are afraid of computers, not for computer experts.
- Norm Matloff's [Chinese-language software page](#).